

Q1.

1.

```
function celsiusToFahrenheit(celsius) {  
  return (celsius * 9/5) + 32;  
}  
console.log(celsiusToFahrenheit(0));  
console.log(celsiusToFahrenheit(12));
```

32

53.6

Explanation: The function applies the formula $F = (C \times 9/5) + 32$ directly. Multiplying by 9/5 scales the Celsius degree size to Fahrenheit's, and adding 32 shifts the zero point (since $0^{\circ}\text{C} = 32^{\circ}\text{F}$, not 0°F).

2.

```
function factorial(number) {  
  let result = 1;  
  
  for (let i = 1; i <= number; i++) {  
    result = result * i;  
  }  
  
  return result;  
}  
  
let number = 9;  
console.log("Number:", number);  
console.log("Factorial:", factorial(number));
```

Number: 9

Factorial: 362880

Explanation

We create a function named factorial(). We start result with 1. A for loop runs from 1 to the given number. Each number is multiplied with result.

3.

```
function isPalindrome(word) {  
  
    let reversedWord = word.split("").reverse().join("");  
  
    return word === reversedWord;  
}  
  
let word = "racecar";  
  
console.log("Word:", word);  
  
console.log("Is Palindrome:", isPalindrome(word));
```

Word: racecar

Is Palindrome: true

Explanation

This code checks whether a word is a palindrome. It reverses the word using split reverse(), and join() then compares it with the original word. If both are the same, it returns true otherwise, it returns false. For example, racecar returns true, while hello returns false.

4.

```
function sumArray(arr) {  
  let sum = 0;  
  for (let i = 0; i < arr.length; i++) {  
    sum += arr[i];  
  }  
  return sum;  
}
```

```
function sumArrayReduce(arr) {  
  return arr.reduce((total, num) => total + num, 0);  
}
```

```
console.log(sumArray([12, 9, 22, 75]));
```

```
console.log(sumArrayReduce([11, 33, 77.9]));
```



118

121.9

Explanation: The loop version accumulates each element into sum. The reduce version does the same thing more compactly. 0 is the starting value, and each call adds the current array element to the running total.

5.

```
for (let i = 1; i <= 15; i++) {  
  if (i % 3 === 0 && i % 5 === 0) {  
    console.log("FizzBuzz");  
  } else if (i % 3 === 0) {  
    console.log("Fizz");  
  } else if (i % 5 === 0) {  
    console.log("Buzz");  
  } else {  
    console.log(i);  
  }  
}
```

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
```

Explanation: For each number from 1 to 15, the code checks divisibility using the modulo operator (%) . It checks "divisible by both 3 and 5" first (otherwise it would print "Fizz" and stop before checking 5), then falls through to individual checks for 3 and 5, and finally prints the number itself if neither applies.

